

MARCO DI PALMA

Software Engineer & Computer Science Student

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PROFESSIONAL SUMMARY

Undergraduate Computer Science student at University of Bari 'Aldo Moro' with strong foundations in low-level systems programming, network architectures, and secure full-stack software development. Passionate about concurrent POSIX sockets, algorithmic efficiency, clean architecture, and building production-grade web systems.

EDUCATION

Bachelor of Science in Computer Science & Software Engineering

University of Bari 'Aldo Moro' (UNIBA) | 2023 - Present

Relevant Coursework: Algorithms & Data Structures, Computer Networks, Operating Systems, Software Engineering, Database Management Systems, Computer Architecture, Object-Oriented Programming.

CORE TECHNICAL COMPETENCIES

- Languages & Systems: C (C99/C11), POSIX Sockets, Multithreading, TypeScript, JavaScript (ES6+), SQL, HTML5/CSS3
- Backend & Web: Node.js, Fastify, React, Next.js concepts, Prisma ORM, RESTful API Design, Clean Architecture
- Systems & Security: Network Protocols (TCP/UDP), I/O Multiplexing, OWASP Hardening, Valgrind, Memory Management
- Tools & DevOps: Git, Docker, Linux/Unix CLI, Vite, Vitest, Doxygen, Postman, Tailwind CSS

KEY SOFTWARE PROJECTS

Weather Station Networking Suite | Low-Level C & Distributed Networking

- Designed a concurrent telemetry architecture in C using POSIX sockets for heterogeneous sensor transmission.
- Implemented UDP datagrams for non-blocking high-throughput ingestion and TCP streams for state synchronization.
- Source: github.com/DarkFury17/weather-station-networking-suite

UniPlan - Intelligent Academic Study Planner | TypeScript, Fastify, React

- Architected a combinatorial exam planner modeling university curricula as Directed Acyclic Graphs (DAG).
- Formulated exam scheduling as a Constraint Satisfaction Problem (CSP) with backward study-load heuristics.
- Built REST API in Fastify with Zod validation schemas, Prisma ORM, Vitest suite (52 tests), and RFC 5545 iCal export.

EnjoyyourCoffee | Production Full-Stack E-Commerce Platform

- Developed a scalable online store with responsive catalog browsing, cart state management, and checkout flows.
- Hardened web security against OWASP vulnerabilities: parameterized SQL queries, session management, CSRF guards.
- Live: enjoyyourcoffee.it | Source: github.com/DarkFury17/EnjoyyourCoffee

Pet Shop Management System | C & Dynamic Data Structures

- Implemented inventory software in C utilizing dynamically allocated linked lists with atomic file persistence.
- Validated 100% memory safety and zero leaks using Valgrind; authored complete technical documentation via Doxygen.
- Source: github.com/DarkFury17/Pet-Shop-Management-System

LANGUAGES

- Italian: Native | English: Professional Working Proficiency (Technical documentation, code review, oral)